Operating Systems (for Yao Class), Fall 2021
General Info

Welcome to the Operating Systems course!

Scope and Topics

The purpose of this course is to teach the design principles of modern operating systems and distributed systems, as well as system programming.

Topics we will cover include concepts of operating systems, networking, database systems and systems programming, including multiple-programing systems (processes, inter-process communication, and synchronization), memory management (segmentation, paging), resource allocation and scheduling, file systems, basic networking (packet switching, file control, reliability), basic databases (transaction, SQL), as well as special topics such as reliability, systems security and practical programming tools. Students are expected to complete set of major design and implementation projects.

Logistics

Time: Thursdays 9:50 – 12:15
Place: 6A303

Credit Units: 4
Course Type: Required Yao-class course
Grading: 100-scale
Exams: One midterm, one final
Expected workload: 12-15 hours per week (including lecture time)
Prerequisites: You should know C++ and basic data structures. Also, you should know basics about computer architecture and assembly languages.

Discussion Session Time: 19:00-20:00 Monday Odd weeks only, place TBD. Optional, but encouraged to attend.

Staff & office hours

Instructor
Dr. Wei Xu 徐葳
Email: weixu@tsinghua.edu.cn
Office Hour: FIT-6004 or at WeChat, after discussion session, or by appointment

Teaching Assistants
Han Zhang 张晗
Office Hour: online at WeChat, or by appointment

Bohan Zhao 赵伯罕
Office Hour: online at WeChat, or by appointment

Grading

- Midterm + final: 40% (20% each)
- Four Projects: 40% (i.e. 10% each)
- Participation / Quizzes: 10%
- Homework 10%
- There will be pop quizzes + cold calls
- Extra points possible for participating in discussions

Textbooks

For the first half semester: *Operating System Concepts (9th Edition)* by Silberschatz, Galvin, and Gagne. (PDF available – ask the TAs)

Second half of the semester - Lecture notes only.

Reading Assignments

The reading assignments for each week are listed in the syllabus page. Except for those marked explicitly with "optional", they are required readings. You are expected to read them before each class. We may have pop quizzes or cold calls in the class to check if you read the materials. Also, all required reading materials will be considered as part of the exam.

Online Teaching Instructions

Unless becoming necessary due to COVID, we will NOT provide online steaming this semester. Everyone is required to attend in the classroom.

Homework and Projects

Practice questions

There are 3 to 4 graded homework assignments for this course. You will need to do it and submit it individually. These assignments are designed to allow you understand the style of the exam questions. We will discuss about the problems during the discussion sessions.

Project format

The best way to learn about operating systems and system programming is building one. You will form teams of 2-3 to complete four (4) non-trivial design and implementation projects.

- Project I: Basics + threads
- Project II: Concurrency
- Project III: Memory management
- Project IV: Storage and RPC

The project is the core of this course, and most of what you learn will be from doing the project. Your homework will be graded automatically against a set of grading rubrics. Copying other peoples' code is strictly prohibited.

Late Policy
We take 10% off the max possible score for each assignment every 24-hour period (or a fraction) after the deadline. There are no exceptions.

To be clear, if the deadline is 11:59pm on October 1, submitting anytime between 12:01am on March 2 – 11:59 on October 2 will receive a 10% penalty, and submitting anytime between 12:01am on October 3 – 11:59pm on October 3 will receive a penalty of 20%, and so on.

The submission time is defined as the timestamp attached by our server and all deadlines are in China Standard Time (UTC+8h) – so please try to submit early to avoid any technical issues!

Team evaluations
Each team will receive the same grade for the projects.

However, we will send you team evaluation forms occasionally to ask everyone to rate each teammates' performance and contribution in the project. If there are significant complaints against a particular team member, we reserve the rights to adjust scores accordingly.

Cheating Policy
We have zero-tolerance on cheating (e.g. copying other people's code or discussions on quizzes). We check code submissions with a state-of-the-art code-copying detector. We report any cheating to the university immediately. Also please protect your own code from being seen by others.

Important event dates
Midterm: week 8
Final: week 16 or 17, TBD